

WP EXAM SUCCESS BOOKING WORKFLOW ENHANCEMENT

Developer Project Specification

1. Project Objective

The existing WP Exam Success WordPress/WooCommerce plugin will be extended so that each booked session is processed independently and the minimum participant check, capacity control, teacher assignment, payment processing and replacement-session process are automated and reliable.

2. Core Development Principle

IMPORTANT: This is an extension of the existing system, not a rebuild. The existing booking, session, WooCommerce, email, cron, timezone and meeting-link infrastructure must be reused. Existing functionality must not be duplicated or redeveloped.

2A. Plugin Independence and Architecture

The WP Exam Success plugin must remain self-contained and independent. New functionality must be implemented within the WP Exam Success plugin and must not rely on theme-specific code, custom theme modifications, Elementor-specific code, or new third-party plugin dependencies, unless an existing dependency is already required by the current plugin architecture.

3. Development Environment and Production Access

- All development, configuration and testing must be performed exclusively on the designated staging website.
- The developer must not make any changes directly to the live/production website.
- The developer must not publish, deploy, migrate or otherwise transfer any changes to the live/production website.
- No access to the live WordPress installation, live database or production hosting environment will be provided to the developer.
- Deployment to the live/production website will be performed exclusively by the client after final acceptance and approval.

- All payment and email testing must use the designated staging/test configuration. No real customer payments or unintended production emails may be used for development or testing.

4. Package and Booking Model

- The customer can choose between three packages: 4 sessions, 8 sessions or 16 sessions.
- The packages have different total prices; the price per session decreases as the number of sessions increases.
- The customer selects the desired dates for the sessions included in the package.
- The selected sessions may take place on different dates.
- The package booking and payment status must be managed independently from the status of its individual sessions.

4. Rotating / Recurring Course System

- The courses are generally ongoing courses that repeat continuously as part of a rotation system.
- New or newly opened courses may also be created.
- Each specific course date/session must be treated independently from other dates/sessions of the same course or rotation.
- At every booking and every teacher-assignment attempt, the current status of the specific course session must be checked again.
- The checks must include the current available capacity and the current teacher-assignment status.

5. Capacity and Teacher Assignment

For every specific session, both participant capacity and teacher assignment must be checked reliably and against the current state.

- Before every booking, the system must verify that at least one place is still available.
- The defined maximum number of participants for the session must never be exceeded.
- Existing reservations must be included in the capacity check.
- Before assigning a teacher, the system must verify whether a teacher has already been assigned to that specific session.
- A session must never be assigned to two teachers at the same time.
- An existing teacher assignment must never be overwritten by a later acceptance.

- Capacity and teacher-assignment checks must remain race-condition safe even when bookings or Accept-Link requests arrive almost simultaneously.
- The existing capacity and reservation logic must be reused and extended for this purpose.

6. Payment Logic

The payment logic must follow the package model. The package price is not charged proportionally per session.

Phase	Required Behaviour
Package purchase	The full package amount is pre-authorized/reserved but not yet captured.
First confirmed session	Once the first session reaches the minimum participant requirement and a teacher has been successfully assigned, the full package amount is captured.
Remaining sessions	All remaining sessions included in the package are covered by the payment already made.
First session does not proceed	No payment is captured. The next successfully confirmed session triggers capture of the full package amount.
A later session does not proceed	That session is not charged separately. The customer's entitlement remains as session credit for a replacement session.

Technical requirement: The developer must verify whether the currently used payment gateway supports a pre-authorization for the period required by the booking process. The existing WooCommerce/payment-gateway integration must be used; no custom payment system may be developed.

7. Required Booking Workflow

Step	Required Behaviour
1. Select package	Existing functionality remains unchanged.
2. Select sessions	Existing functionality remains unchanged.
3. Pre-authorize payment	The full package amount is

	authorized/reserved but not yet captured.
4. Reserve sessions	The existing reservation mechanism is used.
5. Process each session separately	Each booked session is processed independently from the other sessions in the package.
6. Check capacity	Before and during booking confirmation, the system verifies that the session's maximum participant capacity will not be exceeded.
7. Check minimum participants	For each session, the system checks whether at least 5 participants have been reached.
8. Check teacher assignment	The system checks whether the specific session already has an assigned teacher.
9. Assign teacher	When the minimum participant requirement is reached and no teacher is assigned, suitable teachers are automatically invited by email with an Accept Link.
10. Teacher accepts	The first valid teacher to accept is assigned to the session.
11. Confirm session	The individual session is confirmed.
12. Capture package payment	For the first successfully confirmed session, the full pre-authorized package amount is captured.
13. Remaining sessions	Further confirmed sessions are covered by the package already paid for.
14. Session cannot proceed	The customer is informed and must select a replacement session.

8. Independent Session Processing

A package may contain 4, 8 or 16 sessions. Each session must be treated as an independent unit. The outcome of one session must not determine the outcome of the other sessions in the same package.

Session	Result	Consequence
A	Minimum reached; no existing teacher assignment	Teacher assignment → confirmation
B	Minimum not reached	No confirmation → replacement session
C	Minimum reached; teacher already assigned	Session can proceed independently
D	Minimum reached; no existing teacher assignment	Teacher assignment → confirmation

9. Minimum Participant Requirement

- The check is performed separately for each session.
- Global minimum participant requirement: 5 participants.
- The existing Global Settings structure is extended for this purpose.

10. Teacher Assignment via Email and Accept Link

- Once a session reaches the minimum participant requirement and has no assigned teacher, suitable teachers are identified using the existing teacher/class information.
- Suitable teachers automatically receive an email containing a unique, secure Accept Link.
- The Accept Link is valid for 4 hours (changeable).
- Before the actual assignment, the system must check again that the session still has no assigned teacher.
- The first teacher to accept validly is assigned to the session.
- The assignment must be race-condition safe; a session must never be assigned to two teachers.
- If a teacher clicks the link after the session has already been assigned to another teacher, an appropriate message is displayed.
- If no teacher accepts within 4 hours (changeable), the administrator is automatically notified by email.

11. Replacement Session

- A session that cannot take place is handled independently from the other sessions in the package.
- The customer must be informed that the session cannot take place.
- The customer must be offered a replacement session.
- The customer receives a link or other mechanism to select the replacement session.
- The replacement session must not trigger an additional payment because it is covered by the package already purchased.
- The existing session-selection infrastructure must be reused for this purpose.

12. Global Settings

- The existing Global Settings page must be extended to include the required Teacher Assignment settings.
- Minimum Participants: default value 5.
- Automatic Teacher Assignment: ON/OFF.
- Teacher invitation validity: 4 hours. (changeable)
- No separate settings page is required.

13. Existing Infrastructure

The following existing components must be used as the technical foundation and extended only where required for the new workflow:

- WP Exam Success booking and session database
- Existing booking statuses and reservation mechanism
- Existing capacity checking
- WooCommerce order/payment hooks
- Existing WPES_Emailer
- Existing cron infrastructure
- Existing timezone handling
- Existing meeting-link functionality
- Existing customer account / My Sessions
- Existing waitlist infrastructure, where relevant

14. Acceptance Criteria

- Packages with 4, 8 or 16 sessions are processed correctly.
- Multiple sessions within a package are processed independently.
- One session can be confirmed independently of the other sessions in the same package.
- A session below the minimum participant requirement is not confirmed as a regular session.
- At every booking, the system verifies that the maximum participant capacity of the specific session is not exceeded.
- At every teacher-assignment attempt, the system verifies that the specific course session does not already have an assigned teacher.
- Even with almost simultaneous bookings or teacher acceptances, neither the maximum participant capacity may be exceeded nor two teachers assigned to the same session.
- The full package amount is pre-authorized and is captured only when the first session is successfully confirmed.
- If the first selected session does not proceed, no payment is captured.
- The next successfully confirmed session triggers capture of the full package amount in that case.
- Further sessions covered by the already paid package do not trigger additional payment captures.

- A session that cannot take place results in a replacement session without additional payment.
- Suitable teachers automatically receive an invitation with a secure Accept Link.
- The Accept Link expires after 4 hours (changeable).
- Only the first valid teacher acceptance can assign the session.
- A later acceptance cannot overwrite an existing teacher assignment.
- If no teacher accepts within 4 hours (changeable), the administrator receives an email.
- Existing functionality continues to work after the extension without regressions.
- The website must continue to function quickly, smoothly and reliably after implementation. The new functionality must not cause noticeable degradation of loading times, loading problems, errors or instability.
- The plugin must continue to operate independently of the active WordPress theme and must not introduce new dependencies on third-party plugins or theme-specific code. Existing functionality must remain operational when the active theme is changed, provided that existing required plugin dependencies remain active.
- The payment workflow can be fully tested in a safe test/sandbox environment without real payments.
- The plugin is not hard-coded to a specific payment provider and can use a compatible WooCommerce payment gateway configured by the client later.
- The required authorization → later capture workflow is implemented through the WooCommerce payment infrastructure.
- The client can add the final payment-provider credentials and enable the live gateway later without redevelopment of the WP Exam Success plugin.
- All development, configuration and testing has been performed exclusively on the designated staging environment.
- No direct changes have been made to the live/production website by the developer.
- The developer has not published, deployed or otherwise transferred changes to the live/production website.
- The client retains exclusive control of deployment to the live/production website.

15. Delivery and Completion

The project is considered complete when all requirements and acceptance criteria described in this specification have been implemented, integrated into the existing WP Exam Success plugin, tested and delivered without regressions.